

Time Series Correlation Behavior Topology Learning RCA

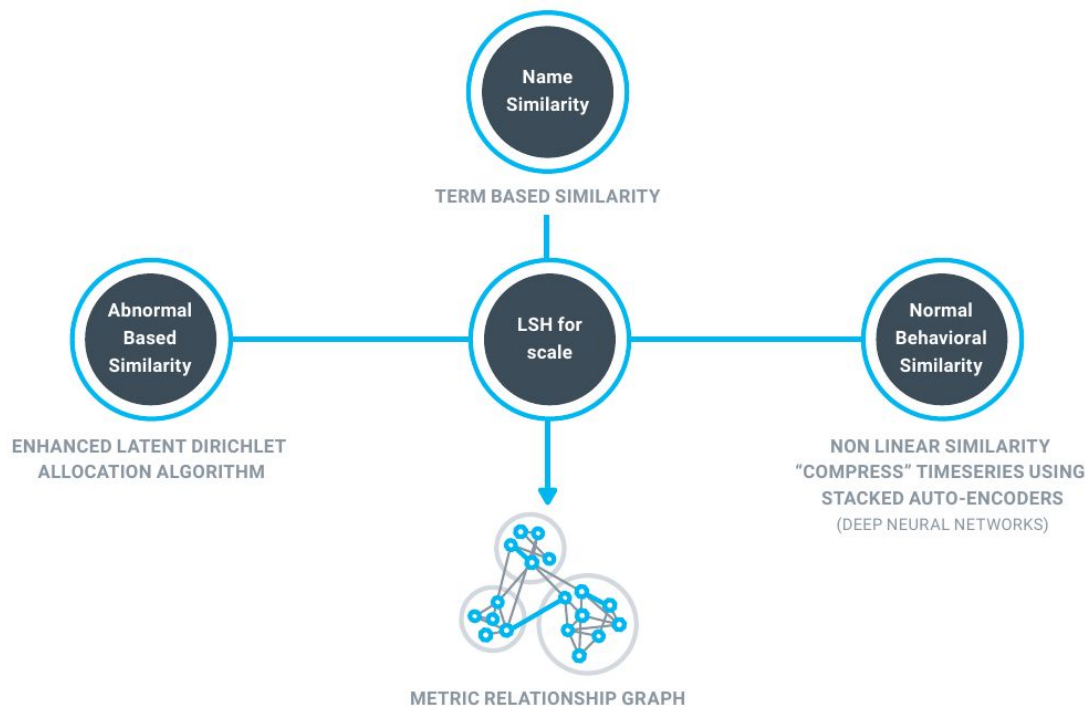


Why look at time series correlation?

- Multivariate anomaly detection:
 - More challenging algorithmically (working in a high-dimensional space).
 - Performance degrades as number of dimensions increases.
 - Difficult to interpret.
- Reduce alert storm, Root cause analysis.
 - Most practically feasible.
- Improve individual detection performance?
 - Will be good if this can be done.
- Good causal model can help with prediction.



Anodot's approach



Methods of relating metrics to each other.



Abnormal behavior based clustering

- Offline data.
- More times that two metrics show anomaly at the same time, the more correlated they are.
- Characterize a metric's time series using 0s and 1s.
 - We could anomaly score.
- Soft clustering based on Latent Dirichlet Allocation.
 - <https://cacm.acm.org/magazines/2012/4/147361-probabilistic-topic-models/fulltext>
- Inspired by FluxInfer, we could consider the time series during the anomalies, also. We have noticed that metrics can get clustered based on how they behave during the anomaly.



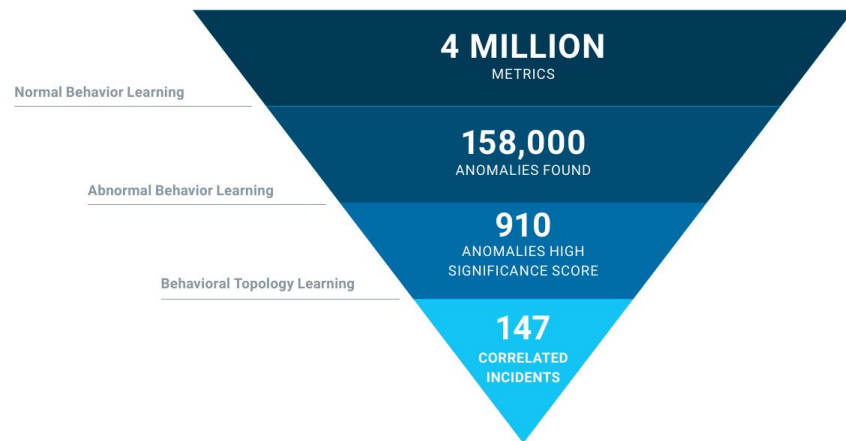
Normal behavior based clustering

- Offline data.
- More challenging than abnormal behavior based clustering.
 - More possibilities about time series behavior.
 - Look hard/long enough, you discover similarities between two time series.
- Pearson correlation.
 - Related to Fisher Z test and RCA literature.
 - Linear: affected by seasonality and trend.
- Look for abstract shapes, instead.
 - Dictionary of shapes: Stacked auto-encoders.
 - TF-IDF to discount common shapes.
 - What is the duration of the shape?
 - Only long periods are mentioned (at least 4 hours).
 - Cannot seasonality and trend affect shape?
 - Especially over a long period of time.



Other techniques

- Name based clustering.
 - Name can be made of terms: term overlap.
- User defined clusters.
- User defined composite metrics.
 - E.g., combine the tables of a database.
- Less data intensive.



About scale

- Anodot repeatedly talks about working with millions and billions of metrics.
 - Is it always with respect to one company?
 - The example is based on a company with 4 million metrics.

Are noise and latent variables too big a challenge?

- Causal analysis: Active topic of research.
- People have been addressing these concerns.
- Causality is as attractive to sell as prediction.
- Go farther in terms of usable results with less labelling burden compared to prediction.
- The time period to be analysed can be marked automatically or manually.



Possible next steps

- Work with Valentin on MAD.
 - There are things to do.

Based on our own experience and discussions with our customers who have faced the “build or buy” decision, we estimate that it would take a minimum of 12 human years (a team of data scientists, developers, UI and QA) to build even the most rudimentary outlier detection system. And this basic system could still encounter various technical issues that are far beyond the scope of this paper.



Possible next steps

- Work with one or two more people and recreate the test set-up in the Microscope paper.
 - Sockshop: Microservices based deployment.
 - Stress-ng CPU saturation; tc based network delay; Container pause.
 - We could try user workload based incidents, also.
 - Will be broadly useful.
- Test Anodot ideas regarding abnormal and normal behavior clustering.
- Test implementations of PC algorithm.
- As a product, these efforts will lead to better RCA.
 - Good UI is important to present these results.



Why is UI important?

- In RCA, conveying the message to the user needs more thought.
- To present the important clusters to look into.
 - In a ranked manner.
- To present causal graphs.
 - To automatically present cause using the causal graph.
- To get user defined description (label) of an incident.
 - Later use these inputs, to present ranked list of causes for a new incident.



PC algorithm packages

- <https://github.com/FenTechSolutions/CausalDiscoveryToolbox>: Python
 - 408 Github stars but paper not mentioning latent variables.
- <https://github.com/microsoft/dowhy>: Python. 2.6k Github stars.
- <https://pypi.org/project/pcalg/>: Python. Paper doesn't mention latent variables.
- <https://www.ccd.pitt.edu/tools/> R, Python, Java.
- <https://stat.ethz.ch/~kalischm/software/>: R package. Mentions latent variables.
- <https://github.com/uber/causalml> 1.6k Github stars.



PC algorithm packages

- <https://mschauer.github.io/CausalInference.jl/latest/>: Julia.
- <https://github.com/philipp-bode/lock-free-pc>: C++.
- Matlab:
 - https://www.cs.utah.edu/~tch/notes/matlab/bnt/docs/usage_cropped.html#structure_learning
 - <http://ofrancois.tuxfamily.org/slp.html>
- <https://github.com/rguo12/awesome-causality-algorithms>
- Causal discovery algorithms: A practical guide by Malinsky and Danks



Nice to know

- What are PacketAI's plans regarding UI?

While one might be tempted to skimp on UI for an internally-developed solution, this is a mistake. An early investment in UI means that the eventual outlier detection system will be able to be used widely in the organization by multiple teams, with multiple needs. If the UI is not simple enough for everyone to learn easily, the data science and business intelligence teams will forever find themselves as the frustrating bottleneck, providing retroactive reports and alerts.

